Research with AI Agents

What Works, What It Costs, and What an Institute Needs

Lukas Theissinger

October 30, 2026

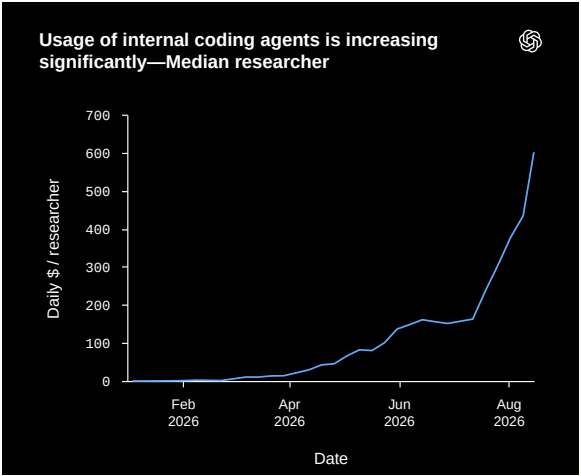
Outline

1. OpenAI and Agent Usage
2. Owning Intelligence and Infrastructure at Lamarr

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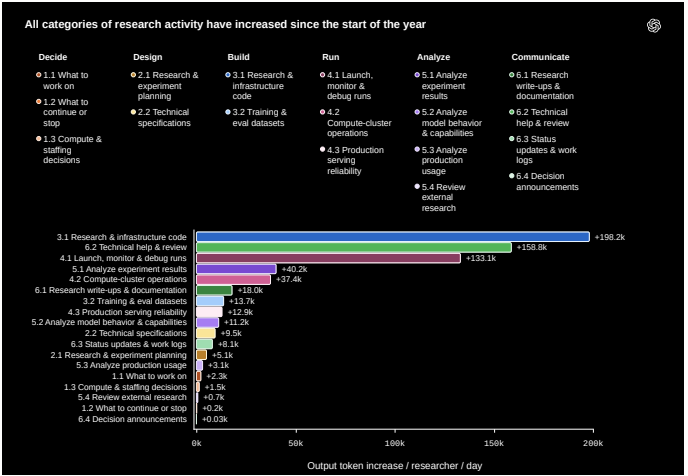
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FOMO Headline: OpenAI Agent Usage



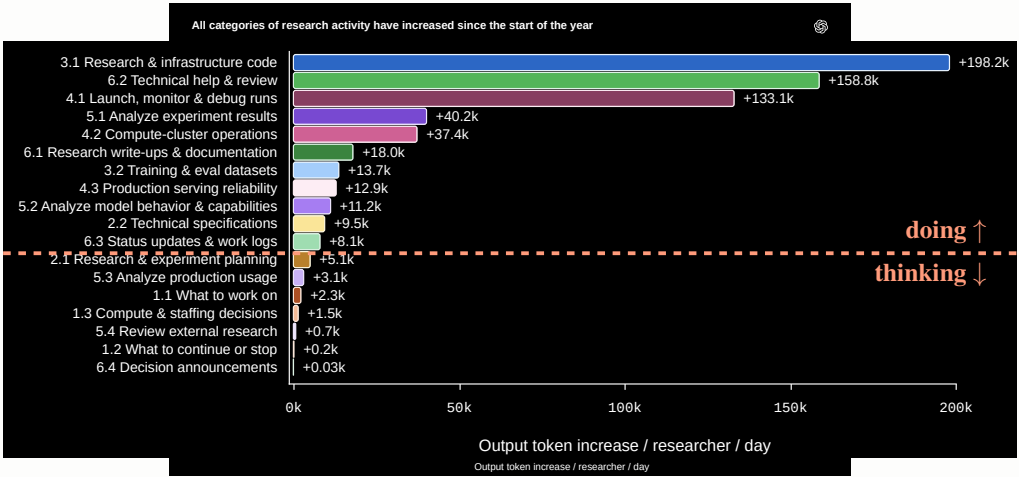
Median researcher, daily $ spent on internal coding agents. Source: OpenAI, “Research acceleration” (self-reported).

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, “Research acceleration” (self-reported).

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What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

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Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure.	Agent use is a *skill*: not everything can or should be automated (degree of automation).

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Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
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4. **Ask your agent to try it:** `github.com/Newida/AutomatingExercises` **Avoid:** Don't do anything BY HAND on this project. If you want to change it, ASK YOUR AGENT to change it.

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Example: travel expense reports

Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** $\times$ **3 h per person per year**. An agent that reads the receipts and files the form correctly needs $\sim$**200 h** (rules, forms, access to the admin system).

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institute, 10 000 people	$10\,000 \times 9\,\text{h} = 90\,000\,\text{h/year}$	200 h	**pays $>$400$\times$**

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Dear management: do not assume that employees will continue spending their own money. We are bottlenecking ourselves: we don't want to spend 100 € per month just to do our work, so we spend 20 €, effectively leaving performance on the table. Also, people will just switch to other institutes, even outside Germany, that supply them with the AI they want and need, for free, to get important work done.

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So let's talk about costs.

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Rule of thumb: people → subscription; pipelines and batch jobs → API.

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Small open models are clearly behind. The largest open models are much closer to the frontier; on our tasks they are **still untested**.

Self-Hosting for the Institute: Pros and Cons

Quality and performance

+ data never leaves Lamarr (privacy, unpublished work, GDPR)
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Rough comparison

$$2 \text{ nodes} \times 300\,\text{k€} \text{ over 4 years} \approx 12.5\,\text{k€/month} \approx \textbf{300 people} \times \textbf{40 €} \approx \textbf{125 people} \times \textbf{100 €}$$

With Lamarr's compute, a frontier-class open model for everyone is realistic: it pays off through privacy, scale and no caps; the closed models keep a lead of months.

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3. **Consider self-hosting.**